

EJECTION PROCESS

SYSTEM BOOT

Box color = active LED color during that phase

USER (MANUAL STEP)

PREPARE FOR EJECTION

1. Load trash into chamber
2. Unlock external hatch safety latch
3. Close & bolt internal hatch

Internal hatch bolted? (limit switch)

If no → button press ignored

PRESS BUTTON

DEPRESSURIZE

Simulated pressure release

OPEN HATCH

Servo opens external hatch

EJECT TRASH

Stepper retracts via rack & pinion · slip gear ejects trash

CLOSE HATCH

REPRESSURIZE

Simulated pressure restore

CYCLE COMPLETE

OLED: "Safe to open"

USER (MANUAL STEP)

RE-LOCK EXTERNAL HATCH

If no further ejections planned

Mid-Cycle Abort

If internal hatch opens or button pressed again → servo stops, red LEDs, OLED "ABORTED" → returns to idle

TIMELINE

SEPTEMBER 2025

Group Formed

- Anyaa & Sophia on airlock; Mohammad & Ian on ejection system
- Defined scope: Mars trash ejection system

NOVEMBER 2025

MVP Created

- VEX + Arduino prototype built
- Limit switch bridged both subsystems
- Full ejection cycle demonstrated

DECEMBER 2025

PDR – Preliminary Design Review

- Presented MVP to judges
- Feedback: add redundancies, one Arduino

JANUARY – FEBRUARY 2026

Refinement + Redundancies

- Added crank, locking pin & limit switch
- Consolidated to single Arduino
- Pareto: focused on highest-impact fixes

MARCH 2026

CDR – Critical Design Review

- Full redundancy stack validated
- Slip-gear mechanism approved by reviewers

APRIL 2026

Final Refinement

- Tuned rack & pinion, 125:1 gear ratio
- Finalized OLED, LEDs & safety interlock
- Poster, website & docs complete

APRIL 2026

Final Design Review

- Full system demo w/ pressure cycle
- All subsystems integrated & verified
- Ready for competition

REDUNDANCIES

CRANK

Manually retracts rack and pinion if stepper fails.

LIMIT SWITCH

Prevents cycle start if internal hatch isn't sealed.

LOCKING PIN

Sliding lever and pin – additional lock beyond the built-in lip.

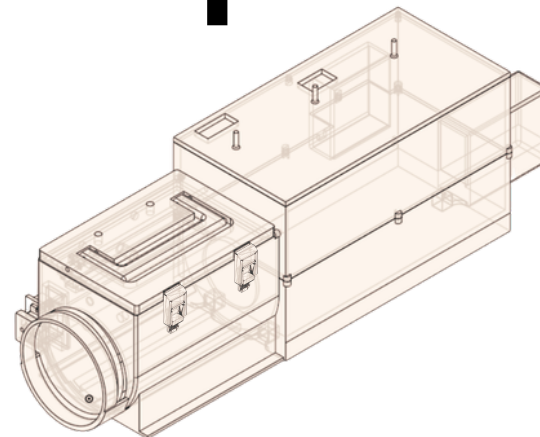
MATERIAL SUGGESTIONS

COMPONENT	MATERIAL	WHY
Chassis	Aluminum 6061	Lightweight, space-proven
Hatches	Titanium	Handles extreme temps
Seals	Silicone Rubber	Airtight in vacuum
Gears & Rack	Stainless Steel	Durable under stress
Electronics	Rad-Hard PCB	Resists Mars radiation
Fasteners	Titanium Bolts	Corrosion resistant

WANT TO WATCH A DEMO? VISIT
HHHWASTEEOFSPACE.WEEBLY.COM



Waste of Space



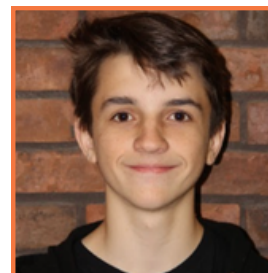
Mars Trash Ejector



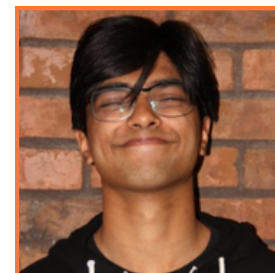
ANYAA SACHDEV



SOPHIA TRAN

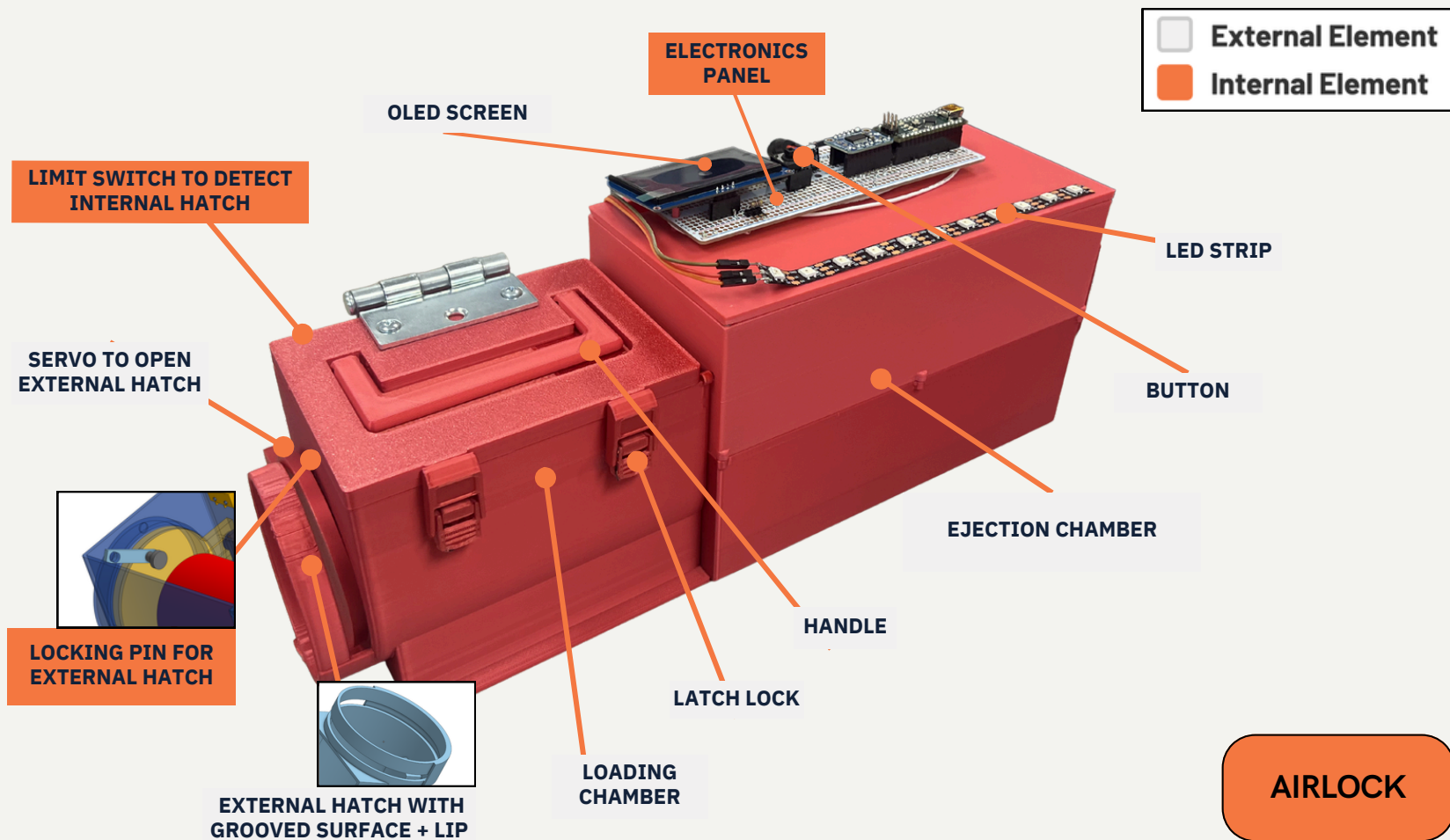


IAN SELIZHUK

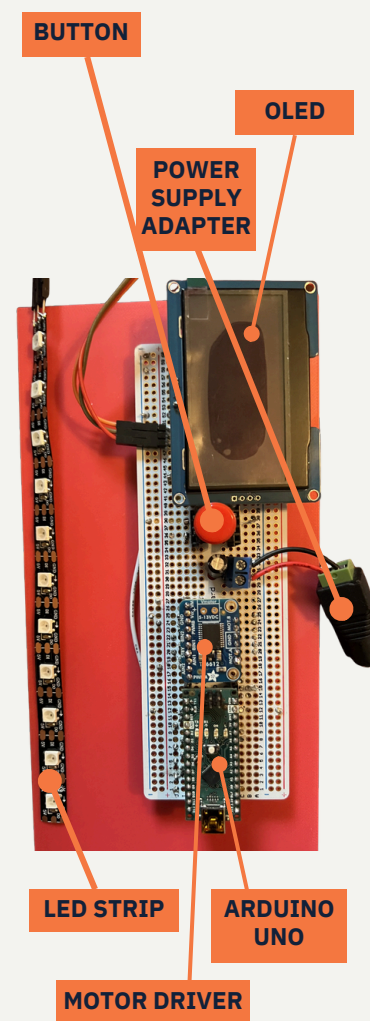


MOHAMMAD WASIM

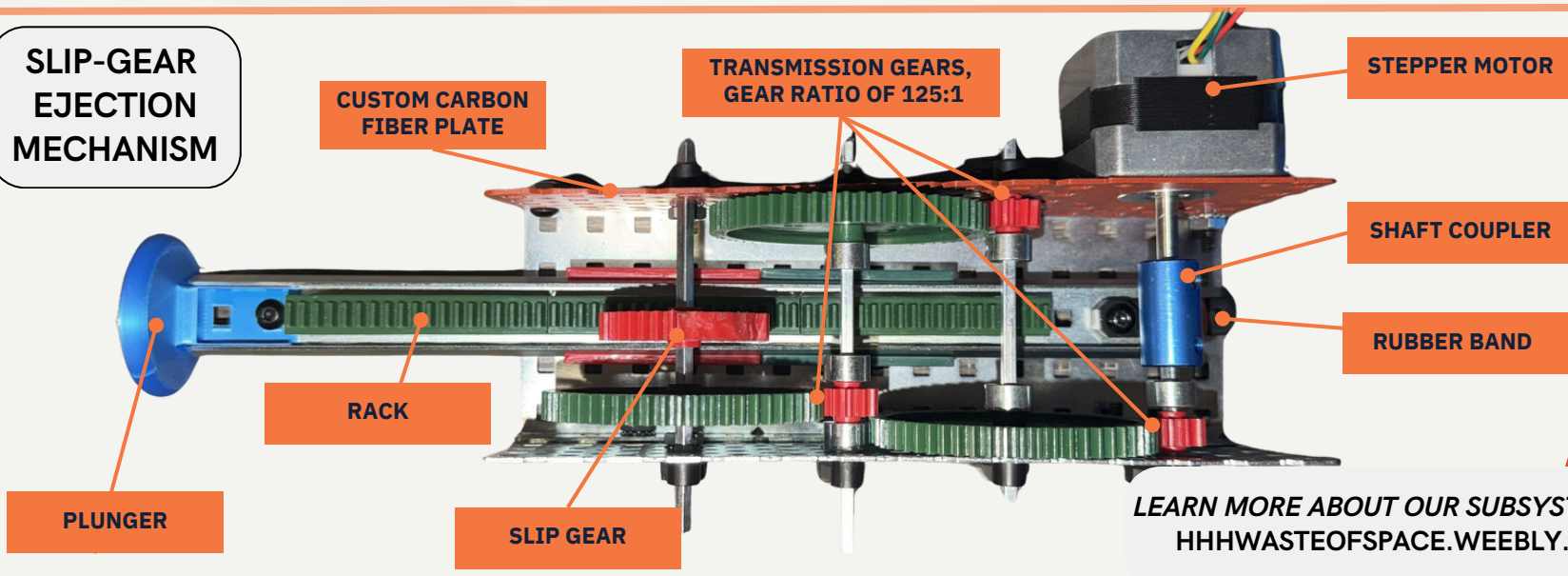
HALF HOLLOW HILLS EAST
MR. REGINI



**ELECTRONICS PANEL
(WITHOUT COVER)
AERIAL VIEW**



**SLIP-GEAR
EJECTION
MECHANISM**



LEARN MORE ABOUT OUR SUBSYSTEMS AT
HHHWASTEOFSPACE.WEBLY.COM

